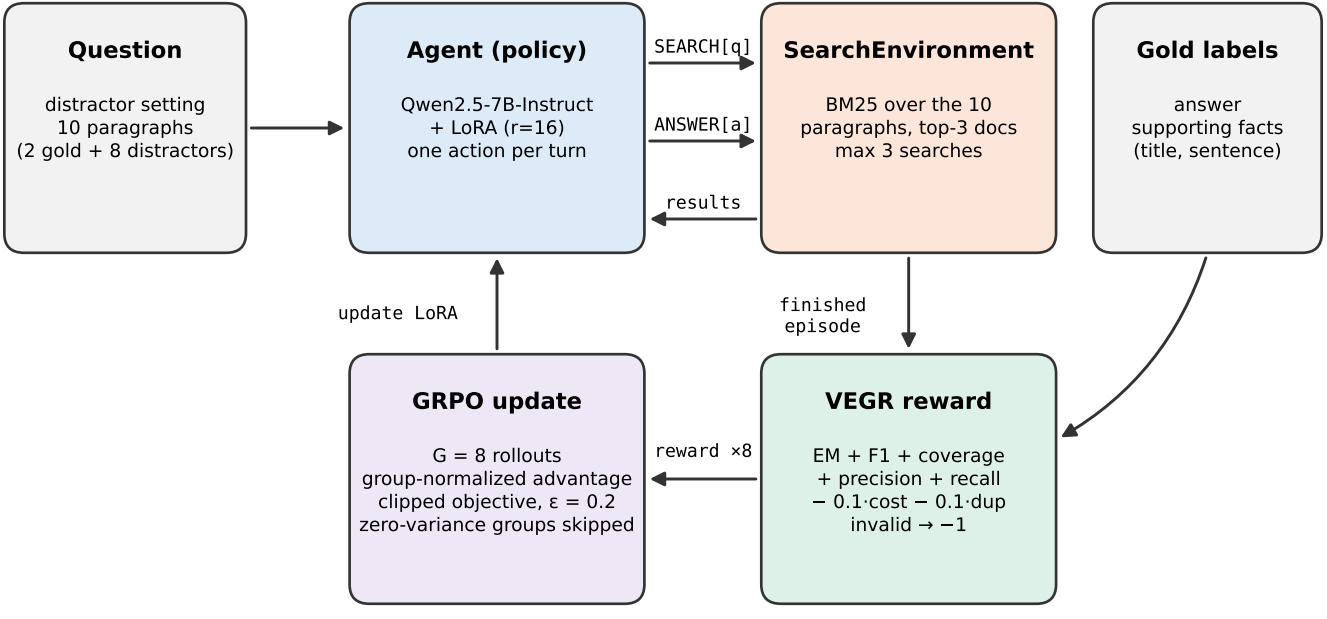
Question
distractor setting
10 paragraphs
(2 gold + 8 distractors)
Agent (policy)
Qwen2.5-7B-Instruct
+ LoRA (r=16)
one action per turn
SEARCH[q]
ANSWER[a]
results
SearchEnvironment
BM25 over the 10
paragraphs, top-3 docs
max 3 searches
Gold labels
answer
supporting facts
(title, sentence)
update LoRA
finished
episode
GRPO update
G = 8 rollouts
group-normalized advantage
clipped objective, ε = 0.2
zero-variance groups skipped
reward ×8
VEGR reward
EM + F1 + coverage
+ precision + recall
− 0.1·cost − 0.1·dup
invalid → −1